Model Report

Pandoc not found. HTML is generated, and Python fallback also exports PDF.

Model Choice

- Selected model: **lasso**
- Selection rationale: Selected lasso as it had the lowest CV RMSE among compared models

Model Comparison

Performance comparison across different models:

Model	CV RMSE (log)	Holdout RMSE (log)	Holdout RMSE ()	*HoldoutMAE*()
---	---	---	---	--- || gradient_boosting	0.1916	0.2136	48561	26442
lasso	0.1439	0.1360	24000	15827
random_forest	0.1507	0.1501	29676	17807

1) Coding (Core Pipeline)

```python
from sklearn.compose import ColumnTransformer
from sklearn.impute import SimpleImputer
from sklearn.linear_model import LassoCV
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import OneHotEncoder, StandardScaler

preprocessor = ColumnTransformer([
    ("num", Pipeline([
        ("imputer", SimpleImputer(strategy="median")),
        ("scaler", StandardScaler()),
    ]), numeric_cols),
    ("cat", Pipeline([
        ("imputer", SimpleImputer(strategy="most_frequent")),
        ("onehot", OneHotEncoder(handle_unknown="ignore")),
    ]), categorical_cols),
])

model = Pipeline([
    ("preprocessor", preprocessor),
    ("lasso", LassoCV(cv=5, random_state=42)),
])
```

```

## 2) Regression Equation

General form:
- y = log1p(SalePrice)
- y_hat = beta0 + sum(beta_j * x_j_tilde)

Expanded equation (Top10 coefficients only):

```text
y_hat = 11.911357 - 0.868460 * cat__RoofMatl_ClyTile - 0.165819 * cat__Condition2_PosN + 0.107589 * num__GrLiv
```

Top10 absolute coefficients:

Rank	Feature	Coefficient
1	cat__RoofMatl_ClyTile	-0.868460
2	cat__Condition2_PosN	-0.165819
3	num__GrLivArea	0.107589
4	cat__Neighborhood_StoneBr	0.098500
5	cat__Neighborhood_Crawfor	0.093396
6	cat__Exterior1st_BrkFace	0.087284
7	num__OverallQual	0.080970
8	cat__Neighborhood_NridgHt	0.068053
9	cat__Functional_Typ	0.058659
10	cat__MSZoning_RM	-0.054916

3) Model Result

- Model type: **lasso**
- Best alpha: **0.00078140**
- Non-zero features: **87**
- Top coefficients/importances file: `outputs/top_coefficients.csv`
- CV protocol: `KFold(n_splits=5, shuffle=True, random_state=42)` with `neg_root_mean_squared_error` on log1p

Top positive coefficients:

Feature	Coefficient
num__GrLivArea	0.107589
```

cat__Neighborhood_StoneBr	0.098500
cat__Neighborhood_Crawfor	0.093396
cat__Exterior1st_BrkFace	0.087284
num__OverallQual	0.080970
cat__Neighborhood_NridgHt	0.068053
cat__Functional_Typ	0.058659
cat__Condition1_Norm	0.048450
cat__BsmtQual_Ex	0.045596
cat__Neighborhood_Somerst	0.043984

Top negative coefficients:

Feature	Coefficient
cat__RoofMatl_ClyTile	-0.868460
cat__Condition2_PosN	-0.165819
cat__MSZoning_RM	-0.054916
cat__CentralAir_N	-0.047431
cat__SaleCondition_Abnorml	-0.046101
cat__Neighborhood_Edwards	-0.042191
cat__LandContour_Bnk	-0.039253
cat__BldgType_Twnhs	-0.033140
cat__BsmtFinType1_Unf	-0.032548
cat__BsmtCond_Fa	-0.026081

Coefficient interpretation
- Numeric features use StandardScaler, so each numeric coefficient means expected change in log1p(SalePrice) f
- Categorical features are one-hot encoded, so each category coefficient is interpreted relative to the omitte
- Rare-category coefficients can be unstable; treat them as predictive signals rather than causal effects.

4) Graph: Predicted vs Actual

`![Holdout Predicted vs Actual](pred_vs_actual.png)`

5) RMSE

RMSE (log1p scale) formula:
- RMSE_log = sqrt((1/n) * sum((y_i - y_hat_i)^2)), where y = log1p(SalePrice).
- This log-scale error emphasizes multiplicative/relative discrepancy.

Back-transform and dollar-scale metrics:
- SalePrice_hat = exp(y_hat) - 1

- SalePrice_true = exp(y_true) - 1
- typical_relative_error = exp(holdout_rmse_log) - 1

Metric	Value
holdout_rmse_log	0.135977
cv_rmse_log_mean	0.143861
holdout_rmse_dollar	23999.75
holdout_mae_dollar	15826.82
typical_relative_error = exp(holdout_rmse_log)-1	0.145656 (14.57%)

- 5-fold CV RMSE (folds): 0.131197, 0.188898, 0.111487

Baseline vs Filtered (Sensitivity Analysis)
- Filtered sensitivity run was unavailable in this execution.
- This is a sensitivity analysis for influential points, not arbitrary deletion of data.
- A better filtered score indicates influence sensitivity; otherwise baseline is already robust.

Appendix: Outlier Diagnostics

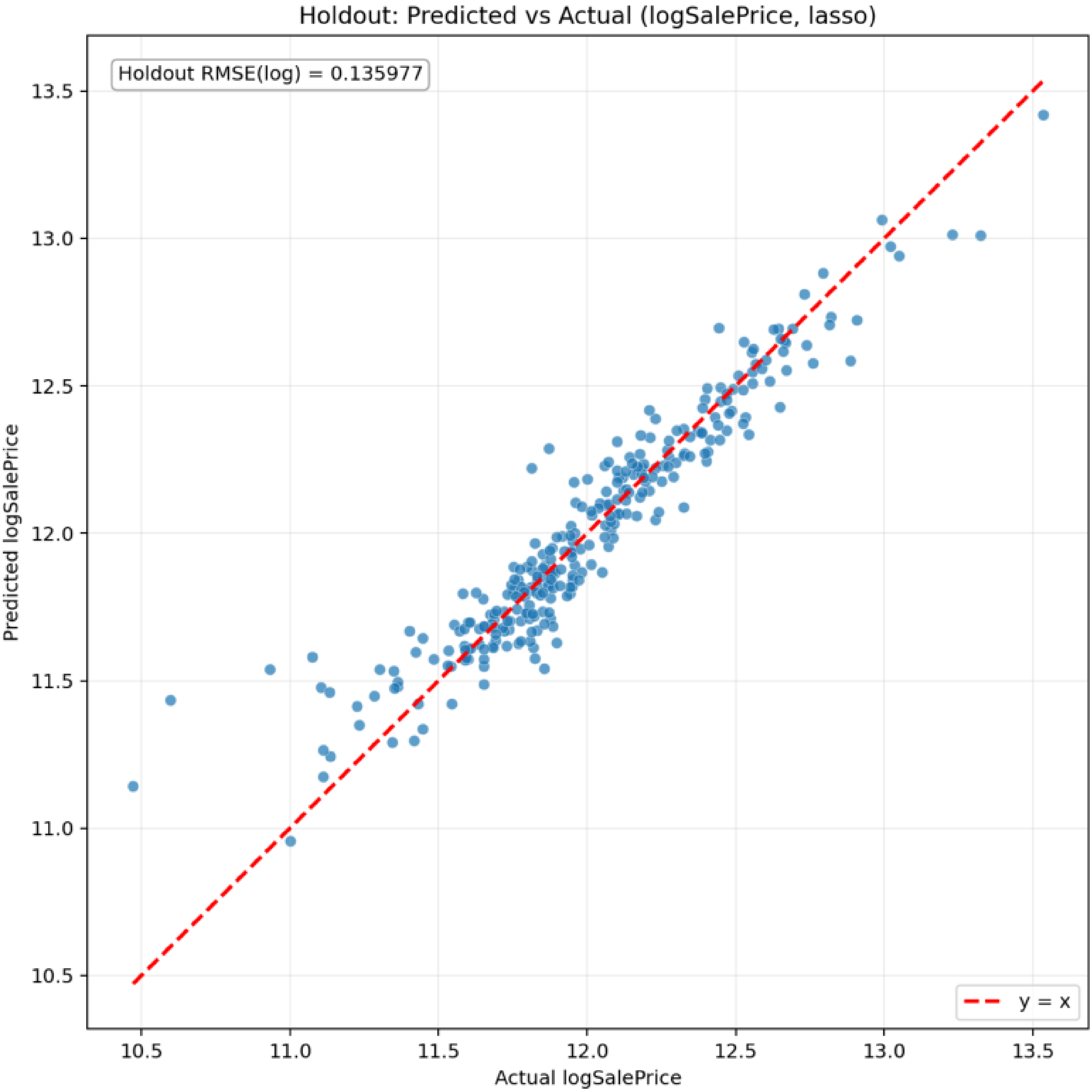
Note: Outlier filtering was not applied in this model comparison run.

Outlier impact figure:
- See `../graph/09_outlier_impact_rmse.png` for RMSE before and after removing high-influence points.
- The chart is a sensitivity check: a noticeable RMSE drop indicates a small set of influential points drives
- If the change is small, model performance is relatively robust to those candidate outliers.

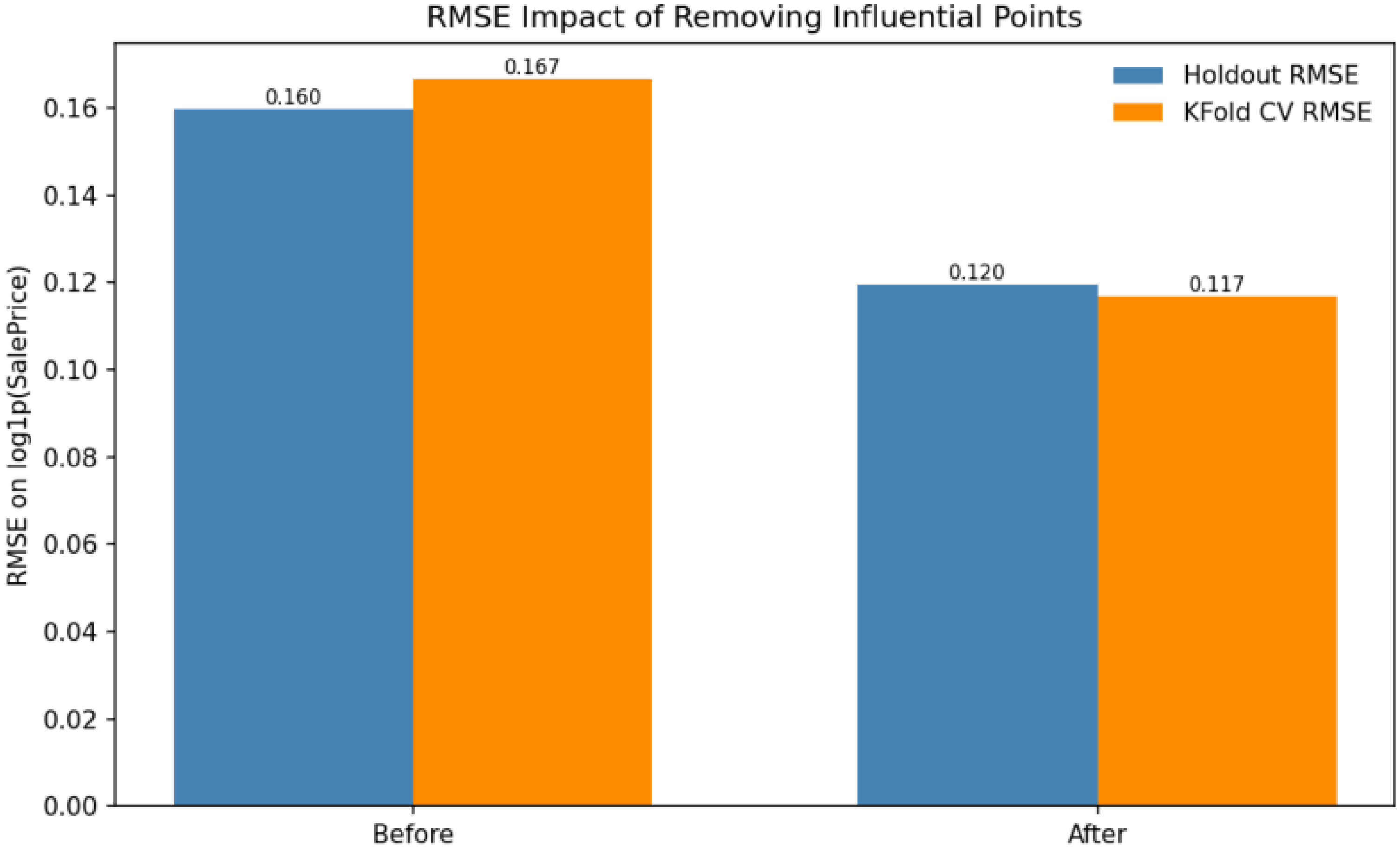
Appendix: Model Comparison Visualizations

Feature importance comparison:
- See `../graph/12_lasso_coefficients.png` for top Lasso coefficients by absolute magnitude.
- See `../graph/13_random_forest_importances.png` for top Random Forest feature importances.
- See `../graph/14_gradient_boosting_importances.png` for top Gradient Boosting feature importances.
- See `../graph/15_model_comparison_rmse.png` for RMSE comparison across all models.
- See `../graph/16_residual_distributions.png` for residual distribution comparison across models.

pred_vs_actual.png



09_outlier_impact_rmse.png



08_cooks_distance.png